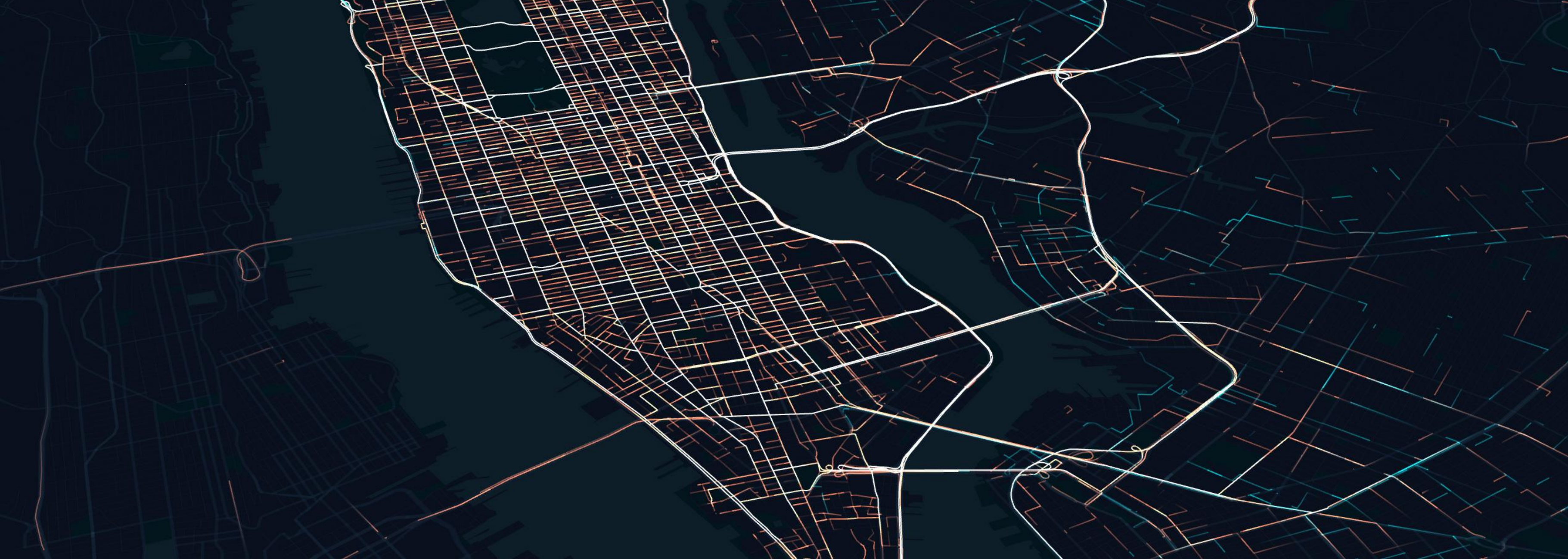




Source: <https://classic.yarnpkg.com/en/package/deck.gl>

Advanced tools and Assignment II

ENVS456 – week 9

Gabriele Filomena

Reflections on Assignment I

- Lectures/Lab sessions are building blocks.
- Data sources (Kaggle, data aggregators, etc.)
- “intermediate maps”, lack of context.
- **Think about interactivity.**
- References.

Towards Assignment II

- **Title:** *Final Geovisualisation Project.*
- **Type:** Coursework.
- **Due: Thursday April 30th (2.00 pm) - Week 11.**
- 50% of the final mark.
- Submission on Canvas, `.html` files only.

This assignment requires you to develop a final geovisualisation project on **spatial data set(s) of your choice**. To be successful, you will need to demonstrate your understanding not only of technical elements, but of the design process required to create a product that can communicate complex ideas effectively. This final product should be a geovisualisation interactive product (i.e., it should contain a cartographic, interactive visualisation) and should contain one analytical component developed further from the content presented in class, in relation to:

- Geographic/Street Network Analysis (week 6).
- Working with Raster Data (week 7).
- Working with spatio-temporal mobility data, not necessarily urban (week 9).

Or, it should present:

- a Dashboard developed with `panel` (week 10) or `pydeck`.



MovingPandas is a Python library for movement data exploration and analysis.

MovingPandas provides trajectory data structures and functions for handling movement data based on Pandas, [GeoPandas](#), and [HoloViz](#).

The official MovingPandas API documentation is hosted on [ReadTheDocs](#).

For more information about individual releases, check out the [Changelog](#).

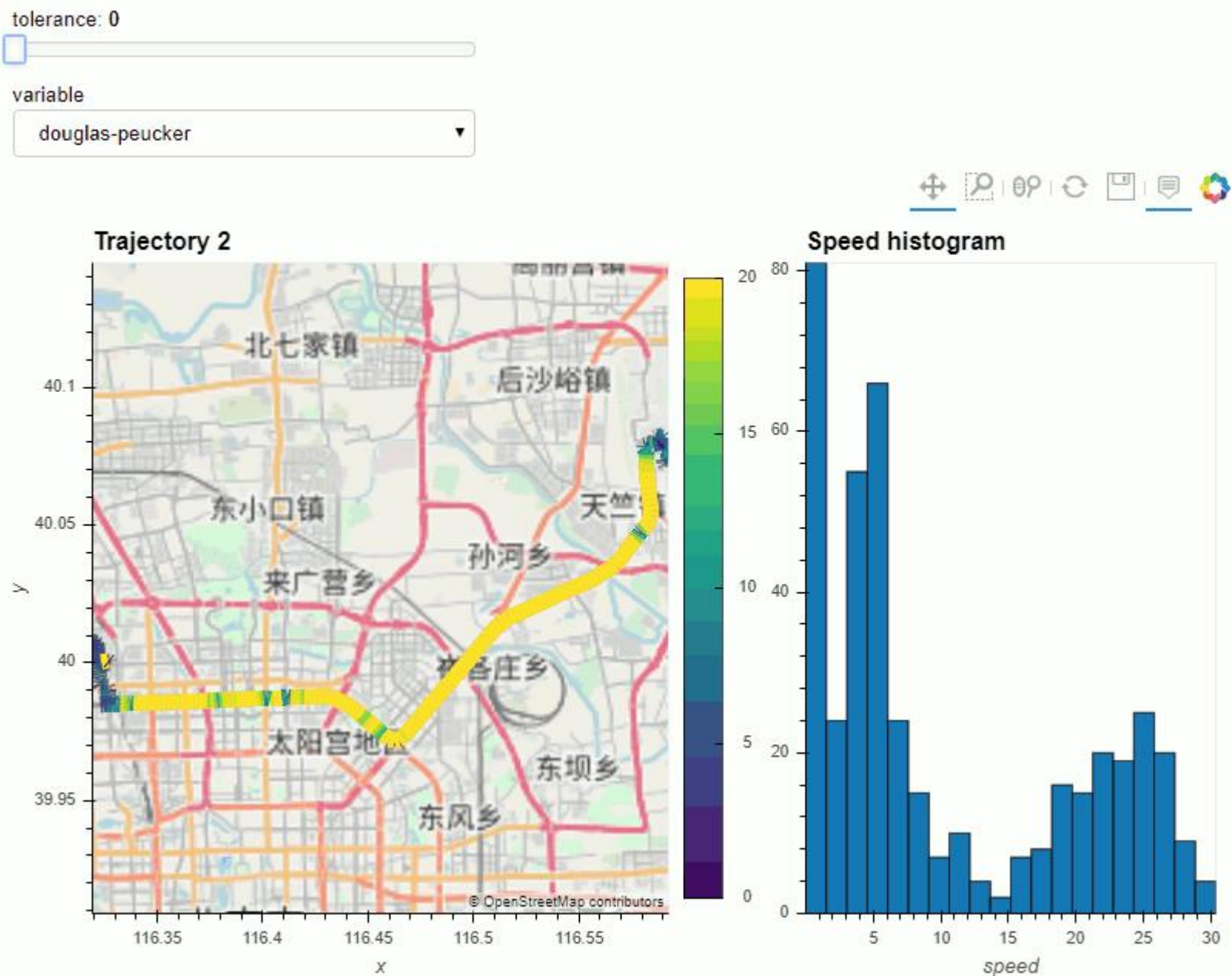
 Follow @movingpandas 338

 docs  passing  PyOpenSci  Peer Reviewed  downloads 87k  launch  binder

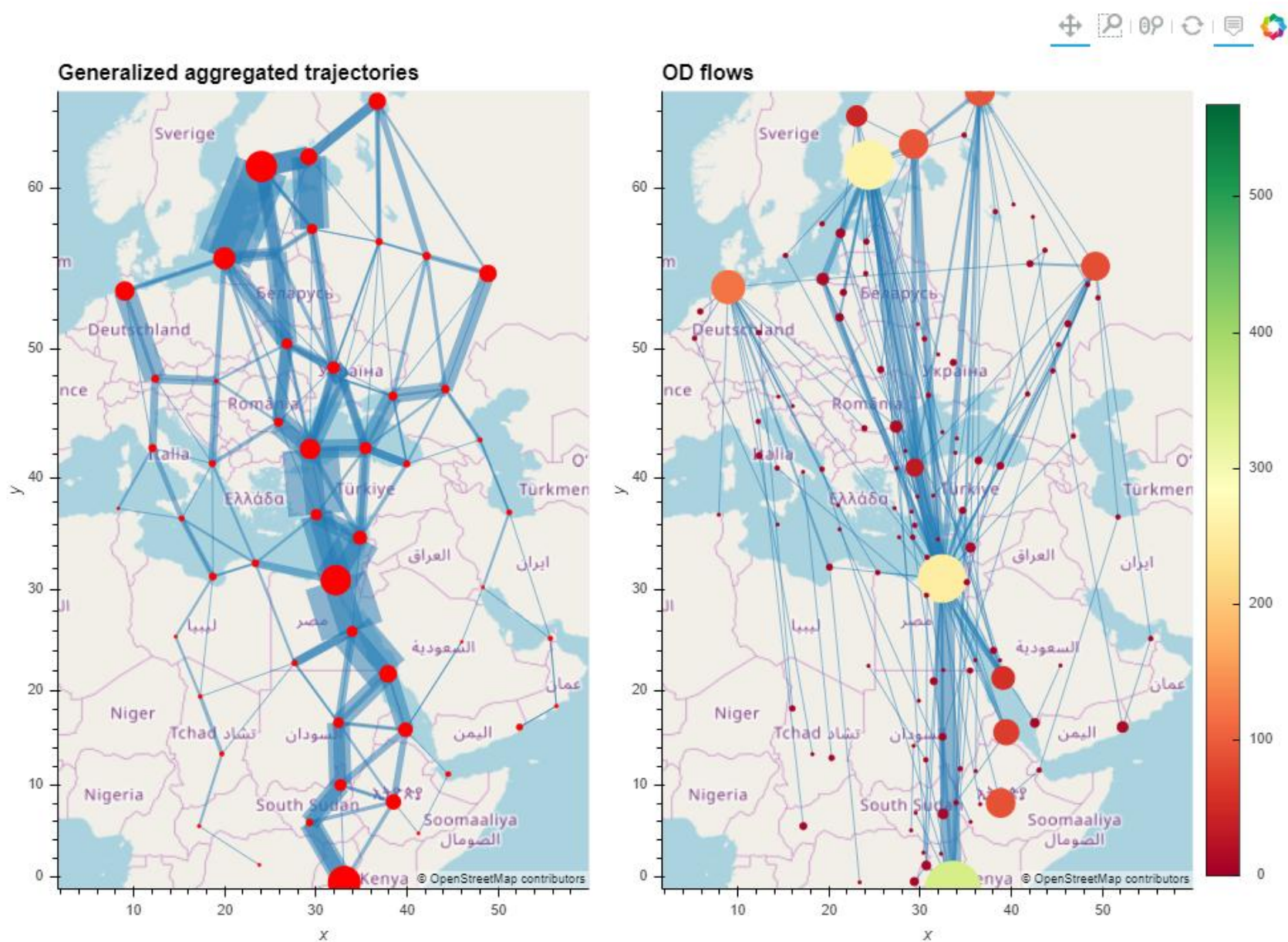
Features

- Easily create trajectories from diverse sources, including CSV files, GIS file formats, (Geo)DataFrames, and OGC Moving Features JSONs (MF-JSON)
- Detect and extract stops
- Split trajectories into individual trips
- Clean, simplify, generalize, and aggregate trajectories
- Find locations for given time stamps and time spans
- Compute movement direction, speed, direction, acceleration, and sampling intervals in a [variety of spatiotemporal units](#)
- Create static and interactive visualizations

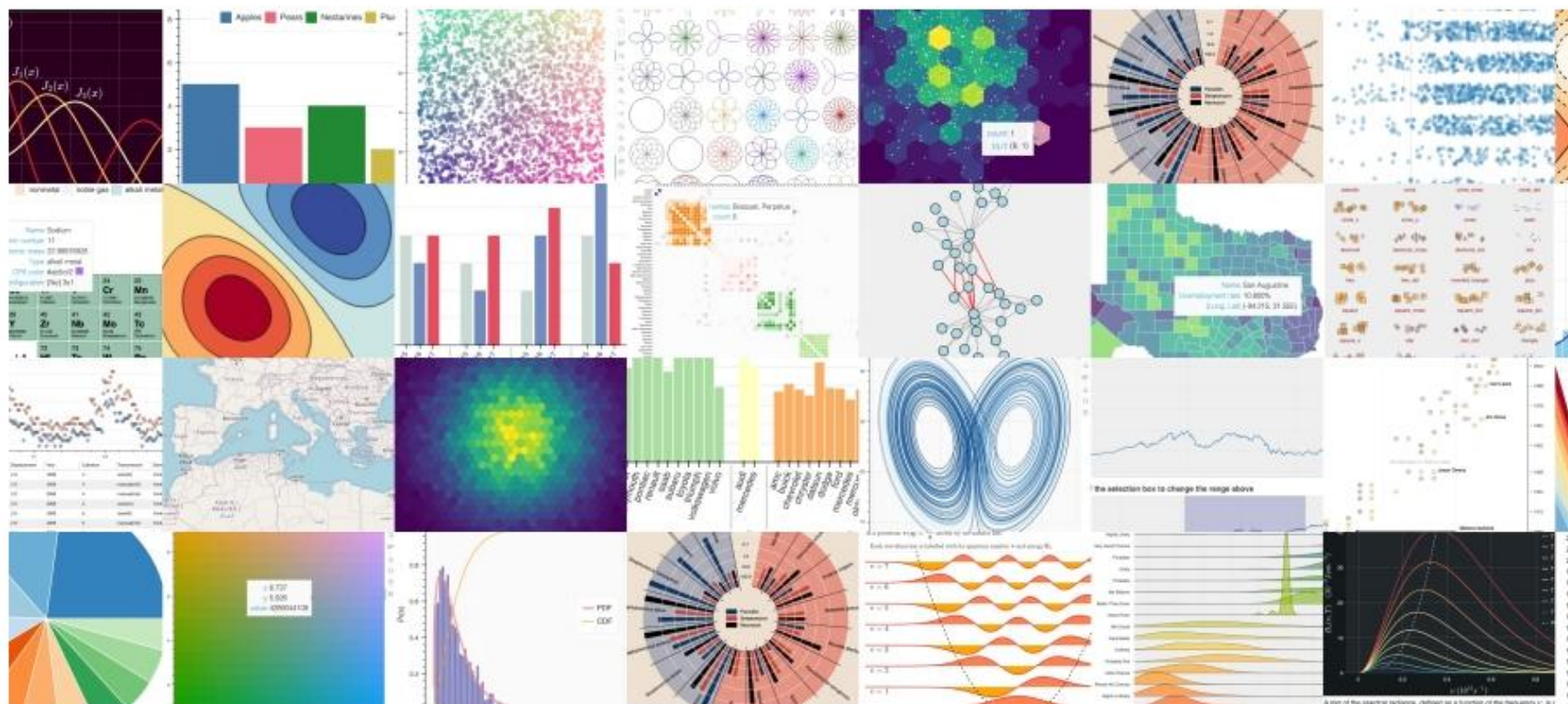
Urban Mobility Analysis



Movement Data Analysis



Other tools

[Documentation](#)[Community](#)[Tutorials](#)[Demos](#)[Blog](#)[GitHub](#)[Donate](#)

Flexible

Bokeh makes it simple to create common plots, but it can also handle custom or specialized use-cases.

Productive

Work in Python close to all the PyData tools you are already familiar with.

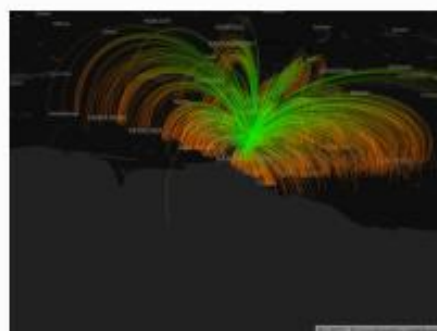


pydeck

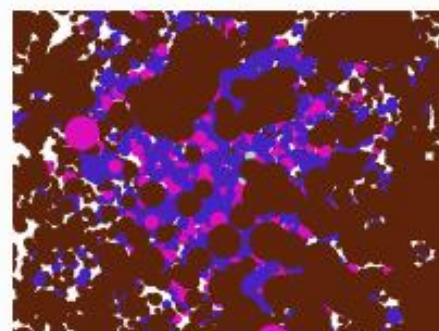
High-scale spatial rendering in Python, powered by [deck.gl](#).

Get started by [installing pydeck](#).

Gallery



ArcLayer



Binary Transport



BitmapLayer



Geographic visualizations for HoloViews

GeoViews is a [Python](#) library that makes it easy to explore and visualize geographical, meteorological, and oceanographic datasets, such as those used in weather, climate, and remote sensing research.

GeoViews is built on the [HoloViews](#) library for building flexible visualizations of multidimensional data. GeoViews adds a family of geographic plot types based on the [Cartopy](#) library, plotted using either the [Matplotlib](#) or [Bokeh](#) packages. With GeoViews, you can now work easily and naturally with large, multidimensional geographic datasets, instantly visualizing any subset or combination of them, while always being able to access the raw data underlying any plot. Here's a simple example:

```
import geoviews as gv
import geoviews.feature as gf
import xarray as xr
from cartopy import crs

gv.extension('bokeh', 'matplotlib')
```



```
(gf.ocean + gf.land + gf.ocean * gf.land * gf.coastline * gf.borders).opts(
    'Feature', projection=crs.Geostationary(), global_extent=True, height=325)
```



Ocean



Land



GeoViews

- Designed to simplify the exploration and visualisation of geographic datasets in Python.
- Supports geographic data handling, including the display and combination of spatial layers.
- Enables interactive spatial analysis, making it easier to explore patterns and relationships.

HoloViz-maintained libraries



pydeck

WebGL-powered visualisation library designed to display large geospatial datasets. A Python wrapper around the JavaScript library *deck.gl*.

- **Rich layered visualisation:** supports a wide range of layers for points, flows, paths, polygons, heatmaps, hexagons, and 3D representations.
- **Interactivity:** supports pan/zoom, hover, tooltips, and other features.
- **Large dataset visualisation:** particularly strong for rendering large spatial datasets efficiently in the browser.
- **Customisation:** detailed control over visual properties such as colour, size, elevation, opacity, and styling logic.

Tool	Best use in practice	Coding power	Main strength	Main limitation	Use case
GeoPandas + Folium	Quick interactive maps to share in notebooks or HTML	High for map personalisation	Easy control of tiles, popups, tooltips, choropleths, GeoJSON styling, layer controls, plugins	Less suitable for very large or visually complex analytical displays	Census map, accessibility map, OD points with popups, interactive coursework deliverable
GeoViews	Exploratory analysis when the data have extra dimensions beyond just geometry	High for exploratory plotting	Strong when combining map layers with dimensions such as time, variable, band, ensemble, and when projection matters	More overhead than Folium	Climate/environmental raster + boundaries + stations; time-varying spatial analysis in notebooks
pydeck	High-performance visualisation of large point, path and flow datasets	High for large spatial datasets	Rich layer catalog and strong rendering for dense data, trajectories, arcs, columns, hexagons, trips	Less convenient than Folium for classic interactive maps	GPS traces, OD flows, large point clouds, mobility dashboards